



Rayane Aboud

Seeking a **work-study (alternance)** contract in France from **September 2026**.
Admitted to **Master 2 Génie Informatique** - Université Grenoble Alpes, France.
Computer Science Engineer, École Nationale Supérieure d'Informatique (ESI), Algeria.

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ABOUT ME

I enjoy applying computer science concepts to optimise real-world systems, with a focus on system design and complex programming languages to improve performance and reliability. I value collaborating with teams on digitalisation projects and engaging with clients to understand their needs and deliver optimal solutions.

Currently I divide my time between managing client solutions at GACA and independently building prototypes inspired by large-scale production environments. Outside work, I enjoy analysing football matches and long walks by the sea.

EDUCATION

Master 2 in Computer Engineering

Université Grenoble Alpes [🔗 Program](#)
Mention Informatique, Parcours Génie Informatique

📅 September 2026 (upcoming, alternance)

- ▶ Coursework focused on building scalable systems and advanced software engineering.

Computer Science Engineer + Master 2 in CS

ESI - École Nationale Supérieure d'Informatique
[🔗 School](#)

Algiers, Algeria

📅 September 2019 – July 2024

- ▶ Specialisation: Computer Systems (Systèmes Informatiques).
- ▶ Coursework: advanced algorithms, networks, cybersecurity, software architecture, queuing theory, simulation, infrastructure, and distributed systems.
- ▶ **Top of class - Mention Très Bien**, GPA 16.27 / 20.

TECHNICAL SKILLS

Interests

Full-Stack Engineering Distributed Systems Optimisation
Java Rust

Backend

Java 8/11 Spring Boot REST APIs Microservices

WORK EXPERIENCE

Optimisation Engineer (Backend Java) - GACA / Avimed

Riyadh, Saudi Arabia (Remote) · Government aeronautical digital services

📅 March 2026 – Present

- ▶ Monitor server performance and maintain clean Java/Spring Boot code practices.
- ▶ Examine system failure points, optimise file organisation, and improve UX/UI based on direct client interaction.
- ▶ Implemented CAPTCHA-based bot validation and ensured frontend compliance with Saudi digital regulations.
- ▶ Build, test, and deploy applications to development and production environments.

Software Engineer - Ouedkniss

Algiers, Algeria · Algeria's largest online marketplace (2 M+ daily users)

📅 July 2024 - February 2026

- ▶ Analysed legacy PHP microservices (2019) and mapped asynchronous service interactions through message queues.
- ▶ Identified costly data bottlenecks and was tasked with rewriting the core revenue-generating service.
- ▶ Redesigned internal components and rewrote the algorithm from PHP to Rust, achieving a **60% reduction in latency** and reducing the operational footprint **from 40 pods to 6**.
- ▶ Built UI tooling to streamline local service execution for the development team; resolved bugs in Rust systems.

INTERNSHIPS

Machine Learning Engineer Intern - Namla

[🔗 Code](#)

Edge Cloud Platform Provider, Algiers · Final-year graduation project

📅 September 2023 – June 2024

- ▶ **Edge:** Handled device authentication, ingested large data volumes, and validated sources/formats via MQTT and NestJS.
- ▶ **Cloud:** Batched and transmitted data via NATS; used Kafka for job queuing and NestJS for processing.
- ▶ **ML:** Trained LSTM models in the cloud (FastAPI/PyTorch) and deployed them to edge inference; tracked performance and generated alerts for abnormal energy consumption.
- ▶ **Frontend:** Built dynamic data visualisation dashboards with React, TypeScript, and InfluxDB.

Network Engineering Intern - ENNA

Entreprise Nationale de la Navigation Aérienne, Algeria · Post-3rd-year internship

📅 Aug 2023 - Sept 2023

- ▶ Explored operations of critical IT systems monitoring Algerian airspace.

Systems & Languages

Rust Go C/C++ Python Bash Linux

Frontend

React TypeScript Vue.js HTML/CSS

DevOps & Cloud

Docker Kubernetes GitLab CI/CD

Messaging

Kafka MQTT NATS Redis

Databases

PostgreSQL InfluxDB SQL

ML / AI

PyTorch LSTM XGBoost FastAPI

Methods

Agile/Scrum TDD Code Review

LANGUAGES

- Arabic Native
- French Fluent (C1)
- English Fluent (C1)
- German Beginner (A1)

PUBLICATIONS

Aboud, R.; Boudaoud, S. A.; Hamdad, L.; Allal, I. (2025). *Edge AI System for Optimising Energy Consumption in Smart Buildings*. Accepted at **MedCCA1 2025**. [DOI](#)

Aboud, R. et al. (2023, internal). *Hybrid Water Wave Optimiser with Simulated Annealing for the Bin Packing Problem*. ESI internal report (French, unpublished). [DOI](#)

COMPETITIONS

Monitoring Data Cup 2025 - Mentor

- Mentored highly competitive students solving real-world AI problems for major Algerian companies (e.g. Yassir).
- Guided competitors in data nature evaluation.

HAICK 2022 - ML Datathon

- School of AI Club, ESI
- Ranked 7th/15 teams; solved industry ML problems using XGBoost and SVM.

- Configured BGP/iBGP routing and integrated ATM systems with a UK aeronautical communication hub via PNETLab and virtual machines.

FREELANCE & STARTUP PROJECTS

Aircrop - Agri-tech Backend (Freelance)

- 2024 - 2025
- Developed the core backend for a startup application managing agricultural fields and generating health reports from drone imagery.
- Built the core infrastructure in Go for high performance, developer ergonomics, and scalability.
- Collaborated with the AI team on health-analysis model deployment and integration.

Aircall Equivalent - VOIP System (Freelance)

- 2026
- Built a VOIP system tailored for the Algerian market.
- Applied advanced system-design principles to implement WebSocket and gateway servers for managing stateful connections.
- Used Rust for secure concurrent connection handling and Vue.js for the frontend.

PERSONAL PROJECTS

Distributed Node Cluster (Rust)

- [Repo](#)
- 2026
- Architected a supervisor-worker cluster over TCP/IP with dynamic node registration, a custom discovery protocol, and real-time membership tracking.

AI-Powered E-commerce App

- [Repo](#)
- 2024
- Integrated the OpenAI API into a Flask app immediately after its public release, enabling natural-language product search where the AI parsed user queries and generated precise database lookups.

C++ Matrix Library

- [Repo](#)
- 2023
- Implemented Householder QR decomposition from scratch, analysing numerical stability, CPU limits, and floating-point constraints; achieves superior stability for matrices larger than 13×13.

ACADEMIC PROJECTS (ESI)

Team Leader - CDN Solution implementation & Deployment

- [Repo](#)
- 2024
- Led a team of 6 to build a Python-based CDN serving educational content (documents, videos) to students nationwide, with nodes in Algiers, Oran, Constantine, and southern regions (TTFB < 200 ms).
- Used HAProxy for load balancing and IP-based session redirection; simulated network infrastructure with OSPF on VMware Workstation.

Team Leader - Bin-Packing Optimisation

- [Code](#) [Report](#)
- 2024

- ▶ Led a team of 5 implementing existing heuristics (tabu search in the repo), then developed a hybrid metaheuristic combining water wave optimisation with simulated annealing (BFS + DFS) in Python - achieving highly competitive results.

Tourism App

 2022

- ▶ Collaborated in a 9-person team to build a tourism application for a client; handled content generation, mapped user trajectories, calculated deployment costs, and pitched the solution as a startup concept.